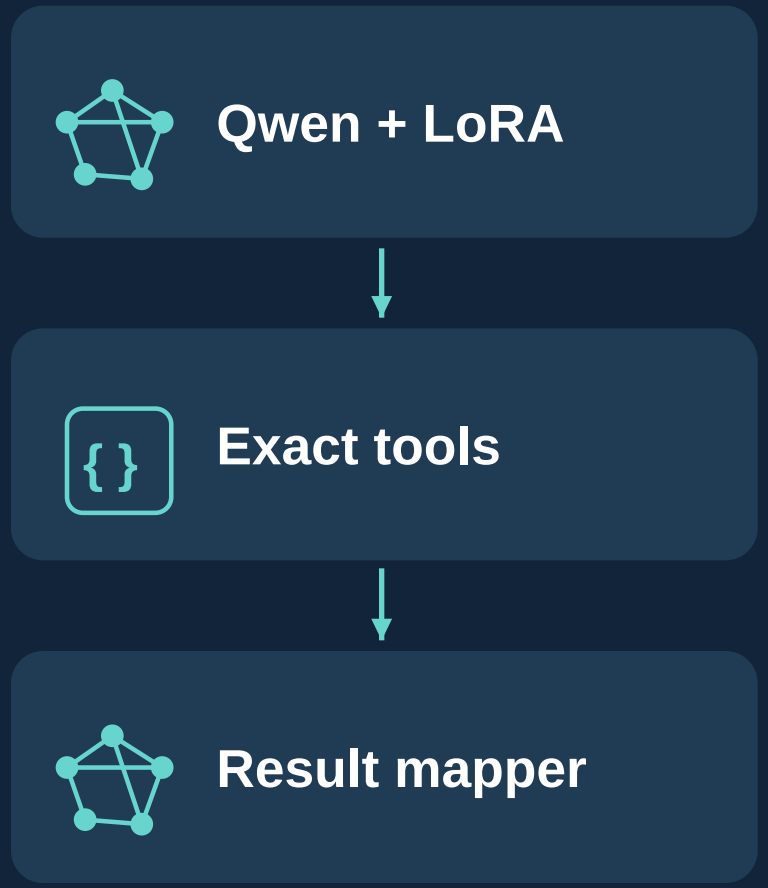
PROJECT REPORT

LLM Character Operations

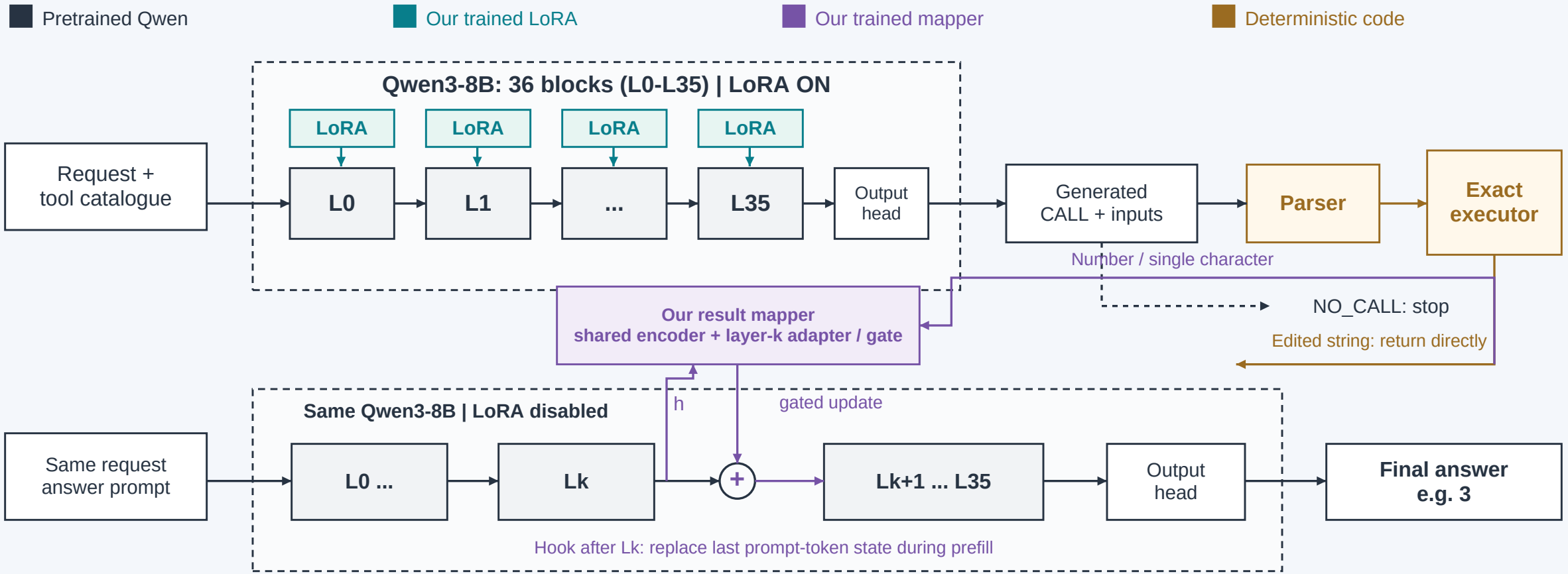
Understand the request.
Compute exactly.
Feed the result back.



Report to Professor | 14 October 2026

Architecture of the inference pipeline

Separate pretrained backbone, trained modules and deterministic computation.

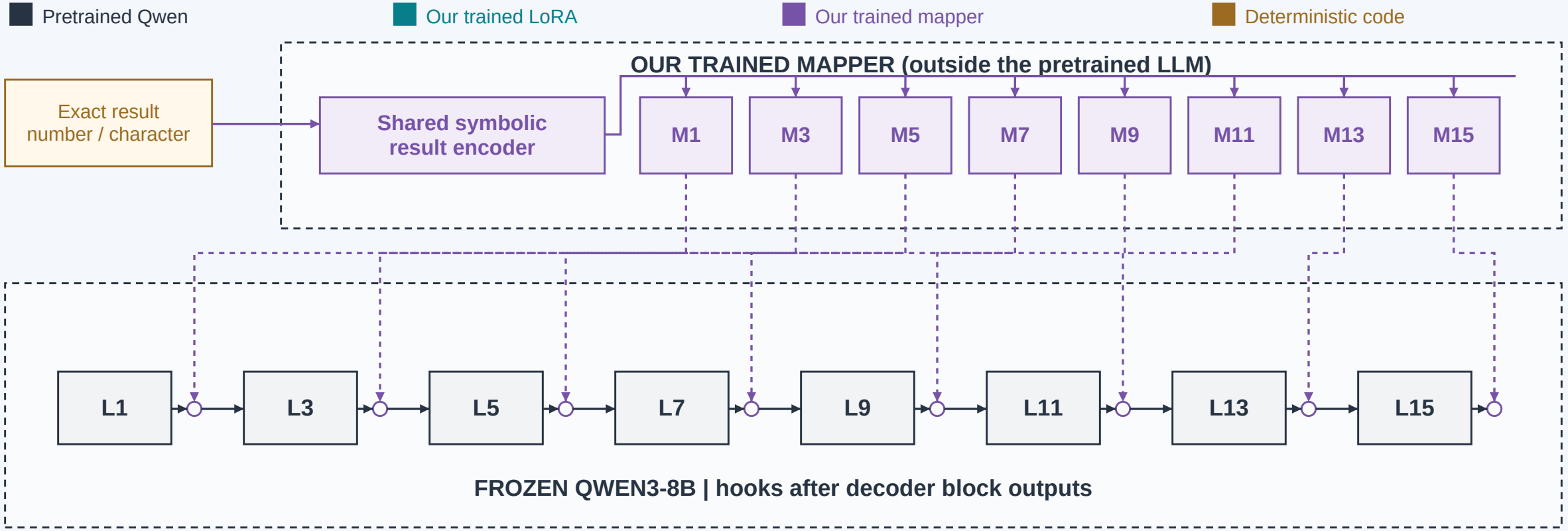


LoRA = small adapters inside Qwen projections, not a separate stack of layers.

Two generation passes. No ground-truth correction of the predicted call.

Layer-wise result injection

Shared result encoding, separate adapters and one selected decoder-output hook.



Even-numbered blocks and later blocks are omitted.

Train across 8 candidates / select on development data / use one hook per answer

Dashed paths are alternatives. Initial three-operation run chose zero-based L5.

Our dataset: CWUM-inspired

Custom English data inspired by Zhang & He's CWUM paper (EMNLP 2024).

50k original / 150k expanded

12 balanced operations

63% tool tasks

37% no-call controls

Count / measure

Characters, text length, words, word length

Find / retrieve

Character position; character in text or word

Edit / reverse

Insert text or words; reverse text or one word

30% real words

50% modified words

20% random strings

CWUM-style prompts + our own templates; exact answers generated by code.

Separate train / dev / test / new-wording test. Expanded data adds longer sentences.

CWUM paper

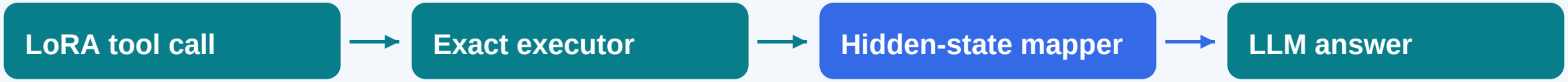
Fresh generated examples; no copied benchmark test items. Tools are allowed in our protocol.

How our pipeline differs

The key change is how a computed number or character returns to the model.

Ordinary tool feedback

Our scalar-result route



Our string-edit route

Same exact tools. Different feedback interface. One call; no retry loop.

We also compare the trained selector with text feedback to assess the mapper's contribution.

Published results and our progress

Accuracy on each study's own benchmark; these rows are not a shared leaderboard.

Study / system	Task / setting	Reported result
CWUM (2024) [1]	English task average	GPT-4: 66.64%
CharBench (2025) [2]	Counting + indexing	GPT-4o: 70.73% overall
Divide & Conquer (2025) [3]	Deletion / insertion / substitution	GPT-3.5: 94.8% / 89.8% / 93.7%
Our initial LoRA + mapper	3 operations; final task answer	IID 99.62% (2334/2343); new wording 98.11% (883/900)
Our newer LoRA selector	12 operations + controls; call exact	Test 99.88%; held-out 94.26% (4713/5000)
Current 12-operation pipeline	LoRA + multilayer-candidate mapper	Completed full-pipeline score not supplied

Initial run: count character, string length, find character; reused V6 mapper at L5. Not the current 12-task protocol.

Initial overall success incl. controls: 99.78% (3991/4000) IID; 96.89% (1744/1800) new wording.

Initial scores verified from copied JSONs; newer selector scores transcribed from server logs.

[1] CWUM, Table 2 [2] CharBench, Table 2 [3] Divide & Conquer, Table 1